

Pb($^{35}\text{Al}, ^{34}\text{Al}n\gamma$) 2017Ch36,2021Bh12

Coulomb dissociation of ^{35}Al on Pb target.

2017Ch36,2014ChZZ: ^{35}Al was produced via the projectile fragmentation of a 531-MeV/nucleon ^{40}Ar primary beam from the Heavy Ion Synchrotron (SIS18) at GSI. The secondary cocktail beam was separated by the FRS separator and impinged on a 2 g/cm² Pb target and a 0.93 g/cm² C target. Projectiles and reaction fragments were detected using 8 DSSDs, separated by a large-area dipole magnet (ALADIN) and tracked using two large scintillator fiber detectors (GFIs). Neutrons from the excited projectiles were detected using the high-efficiency Large Area Neutron Detector (LAND). γ rays from the deexcited projectile and projectile-like fragments were detected using a spherical 4 π Crystal Ball detector array of 162 NaI(Tl) crystals. Measured $E(\text{fragment})$, E_n , E_γ , Coulomb dissociation cross sections. Deduced relative populations of ^{34}Al , ^{35}Al g.s. configuration. Comparisons with shell-model calculations with the SDPF-M interaction. The measured inclusive differential CD cross section (integrated up to 5.0 MeV relative energy) for $^{35}\text{Al} \rightarrow ^{34}\text{Al}+n$ using a Pb target is 78 mb *I3*.

2021Bh12: A further analysis of the data from **2017Ch36**. The $^{35}\text{Al}(\gamma,n)^{34}\text{Al}$ photoabsorption cross section was obtained from fitting the direct breakup model to the measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al core excited states. The $^{34}\text{Al}(n,\gamma)^{35}\text{Al}$ neutron capture cross sections were obtained from the photoabsorption cross sections using the detailed balance theorem.

 ^{35}Al Levels

<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0	(5/2 ⁺ , 3/2 ⁺ , 1/2 ⁺)	J^π: From comparisons of measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al in its g.s. and/or 46-keV isomer with theoretical calculations from the direct breakup model using the plane-wave approximation assuming the valence neutron at different orbitals. 2017Ch36 states that the differential CD cross section of $^{35}\text{Al} \rightarrow ^{34}\text{Al}+n$ has been interpreted in the light of a direct breakup model, and it suggests that the possible ground-state spin and parity of ^{35}Al could be, tentatively, 1/2⁺ or 3/2⁺ or 5/2⁺. Major configurations and spectroscopic factor for $J^\pi=5/2^+$ of ^{35}Al g.s.: (g.s., 4⁻ in ^{34}Al)$\otimes$$\nu p_{3/2}$, $S=0.36$ 9 (2017Ch36); (46 keV, 1⁺ in ^{34}Al)$\otimes$$\nu d_{3/2}$, $S=1.47$ 22 (2017Ch36); (1.4 MeV, 2⁺ in ^{34}Al)$\otimes$$\nu s_{1/2}$, $S=0.16$ 1 (2021Bh12); (2.5 MeV, 3⁻ in ^{34}Al)$\otimes$$\nu p_{3/2}$, $S=1.48$ 18 (2021Bh12). Other configurations for $J^\pi=1/2^+, 3/2^+$ of ^{35}Al g.s.: (g.s., 4⁻ in ^{34}Al)$\otimes$$\nu f_{7/2}$, $S=1.03$ 43 and (46 keV, 1⁺ in ^{34}Al)$\otimes$$\nu s_{1/2}$, $S=0.62$ 7; (46 keV, 1⁺ in ^{34}Al)$\otimes$$\nu s_{1/2}$, $S=0.72$ 8; (46 keV, 1⁺ in ^{34}Al)$\otimes$$\nu s_{1/2}$, $S=0.45$ 7 and (46 keV, 1⁺ in ^{34}Al)$\otimes$$\nu d_{5/2}$, $S=0.94$ 22.